

Self-organizing optic-cup morphogenesis in three-dimensional culture

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Balanced organogenesis requires the orchestration of multiple cellular interactions to create the collective cell behaviours that progressively shape developing tissues. It is currently unclear how individual, localized parts are able to coordinate with each other to develop a whole organ shape. Here we report the dynamic, autonomous formation of the optic cup (retinal primordium) structure from a three-dimensional culture of mouse embryonic stem cell aggregates. Embryonic-stem-cell-derived retinal epithelium spontaneously formed hemispherical epithelial vesicles that became patterned along their proximal–distal axis. Whereas the proximal portion differentiated into mechanically rigid pigment epithelium, the flexible distal portion progressively folded inward to form a shape reminiscent of the embryonic optic cup, exhibited interkinetic nuclear migration and generated stratified neural retinal tissue, as seen *in vivo*. We demonstrate that optic-cup morphogenesis in this simple cell culture depends on an intrinsic self-organizing program involving stepwise and domain-specific regulation of local epithelial properties.

Eye formation has attracted the attention of both classical and modern-day developmental biologists^{1–12}. The retinal anlage, which is demarcated by Rx (also called Rax) expression^{13,14}, first appears as the optic vesicle, an epithelial vesicle evaginating laterally from the diencephalon. Subsequently, its distal portion invaginates to form a two-walled cup-like structure, the optic cup (Supplementary Fig. 1a), which develops into the outer (pigmented) and inner (neurosensory) layers of the retina. Optic-cup development occurs in a complex environment affected by many neighbouring tissues. Therefore, its mechanistic analysis is far from simple, and such complexities have led to controversial results^{9,10}, in particular, with regard to the requirement of the surface ectoderm and lens in the invagination of the neural retina^{15–18}.

Here, we took an experimental approach using a three-dimensional (3D) embryonic stem (ES) cell culture system^{19,20} and successfully reduced the complexity of this organogenetic process.

Optic-cup self-formation in 3D ES cell culture

We previously demonstrated the self-formation of stratified cerebral cortical tissues in culture, where floating aggregates of ES cells were cultured under low growth-factor conditions (serum-free floating culture of embryoid-body-like aggregates with quick reaggregation or SFEBq)^{20,21}. We could also induce retinal differentiation in a modified SFEBq culture using transient activin treatment²², but no clear formation of retinal epithelial structures was observed (data not shown). We then further modified the culture medium (see Methods) and added basement-membrane matrix components (matrigel) to promote the formation of rigid continuous epithelial structure²³ (Supplementary Fig. 1b–d). These modifications successfully improved the efficiency of retinal induction, resulting in 30–70% Rx-GFP⁺ cells in total cells (typically, ~80% of aggregates were strongly positive for Rx-GFP; see Supplementary Fig. 1e for the dependence on integrin signals). A similar high percentage of Rx induction was induced by treating ES cell aggregates with defined matrix proteins (purified laminin and

entactin) in the presence of the activin-family protein Nodal during days 1–7 (Supplementary Fig. 1f).

By day 6, initially homogenous ES cell aggregates formed hollowed spheres consisting of polarized N-cadherin⁺ neuroepithelium with the apical surface inside (Supplementary Fig. 1g–j), which spontaneously subdivided into Rx-GFP⁺ and Rx-GFP[−] portions (Fig. 1a and Supplementary Movie 1). On day 7, the Rx-GFP⁺ portions formed hemispherical epithelial vesicles evaginating from the main body with one to four vesicles per aggregate ($73.7 \pm 2.5\%$ Rx⁺ spheres; Fig. 1b, c). The formed Rx-GFP⁺ tissues were Pax6⁺ Sox1[−] (Fig. 1d, e), consistent with retinal marker expression^{19,22}, whereas the Rx-GFP[−] portions were the non-retinal neuroectodermal epithelium (Sox1⁺; Fig. 1e).

Notably, on days 8–10, the Rx-GFP⁺ vesicles underwent a dynamic shape change and formed a two-walled cup-like morphology ($57.0 \pm 4.7\%$ of vesicles, Fig. 1f, g and Supplementary Fig. 1k, l; typically 200–400 μm in diameter; the optic cup in the E10.5 mouse is ~300 μm). The distal portion of the vesicle invaginated and expressed neural retina markers^{13,14,24} such as Chx10 (also called Vsx2) and Six3 in addition to Rx and Pax6 (Fig. 1h, i and not shown). In contrast, the proximal epithelium, now forming the outer shell, expressed Pax6, Mitf, Coup-TF2 (also called Nr2f2), Otx2 and connexin 43, and a low level of Rx (Fig. 1i, j and Supplementary Fig. 1m–p), reminiscent of the marker profile of retinal pigment epithelium (RPE) progenitors^{4,25}. This outer portion subsequently became pigmented (Fig. 1k). Like the one *in vivo*, the ES-cell-derived optic cup had a clear apical (aPKC)–basal (laminin) polarity (Fig. 1l), and the inner portion invaginated with its apical side convex.

Importantly, the formation of the optic cup in the ES cell culture occurred in the absence of a lens (no crystalline⁺ tissue; $n = 50$ aggregates) or surface ectodermal tissues (see the *in vivo* situation in Fig. 1m and Supplementary Fig. 1q; compare to Fig. 1h), indicating that the invagination was not caused by external structures but occurred in a self-directed fashion (Fig. 1n). Hydrostatic pressure seems to have no

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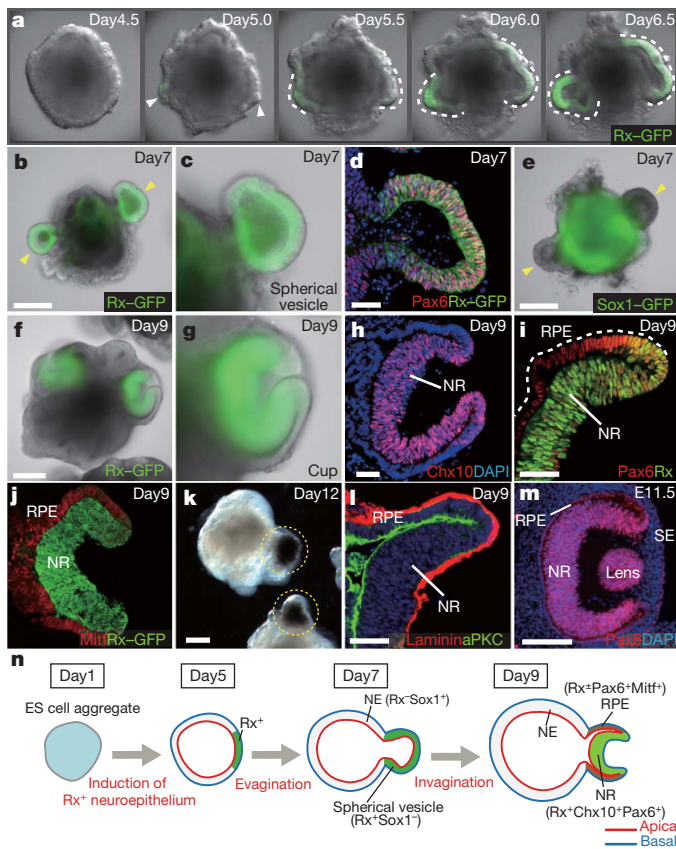


Figure 1 | Self-formation of an optic-cup-like structure in 3D culture of ES cell aggregates. **a–e**, SFEBq/matrixgel culture. **a–e**, Self-formation of vesicles expressing Rx-GFP (**a–c**) and Pax6 (**d**), but not Sox1-GFP (**e**). **f–i**, Rx-GFP⁺ eye-cup structures on day 9 (strongly and weakly in the neural retina (NR) and RPE portions, respectively). **h**, **i**, The inner portion strongly expressed Chx10 (**h**) and Rx and moderately expressed Pax6 (**i**) on day 9. **j**, **k**, The outer epithelial shell of the cup expressed Mitf (**j**; day 9) and accumulated pigment on day 12 (**k**). **l**, The apical marker aPKC and laminin⁺ basement membrane. **m**, E11.5 mouse eye. **n**, Schematic of optic-cup self-formation. SE, surface ectoderm. Scale bars: 200 μ m (**b**, **e–f**, **k**), 50 μ m (**d**, **h**, **i**, **l**), 100 μ m (**m**).

essential role, as opening a small hole through the neuroectodermal epithelium did not substantially affect eye-cup formation.

Four phases of 3D eye-cup morphogenesis

We then continued to analyse the morphogenetic process in 3D, using a specially assembled multi-photon live-imaging system in combination with a full-sized CO₂/O₂ incubator (Fig. 2a and Supplementary Fig. 2a, b), which allows a constant, healthy culture environment for a week or longer (see Fig. 2b and Supplementary Movie 2, part a, for an example of the time-lapse deep imaging). The invagination process consisted of four consecutive phases (Fig. 2c, d and Supplementary Movie 2, parts b, c), consistent with *in vivo* development²⁶. On day 6, the evaginated vesicle was hemispherical in shape (phase 1). In phase 2, occurring around day 7, the distal portion of the vesicle became flattened (Supplementary Fig. 2c). Subsequently, in phase 3, the angle at the joint (hinge) between the neural retina and RPE domains became narrower or even acute (Fig. 2e). Then, on day 8, the neural retina epithelium started to expand as an apically convex structure, forming a cup via progressive invagination (phase 4).

The microfilament system has been implicated as a key internal regulator in various aspects of epithelial morphogenesis and is often regulated by the Rho-ROCK system²⁷. The invagination morphogenesis (including flattening and hinge formation) was blocked when the culture was treated with the ROCK inhibitor Y-27632 (ref. 28) (which is known to attenuate myosin activity) from phase 1 or 2 (Fig. 2f and data

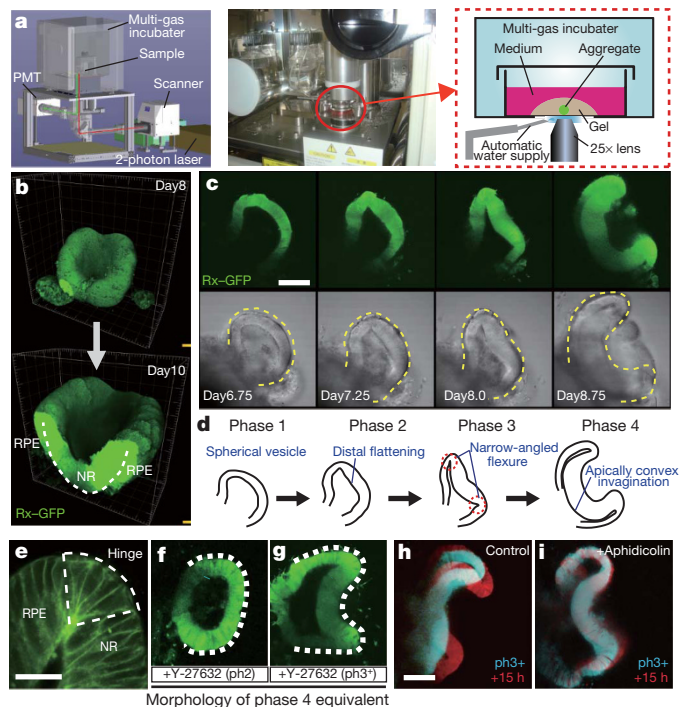


Figure 2 | Progressive morphogenetic changes of ES-cell-derived retinal epithelium. **a**, Multi-photon device for long-term 3D live imaging. PMT, photomultiplier. **b**, Surface-rendering 3D reconstruction images of invagination. The front part (bright green) is a cross-section image. **c**, Optical cross-section of Rx-GFP images (top) and laser-scanned bright-field images (bottom; dotted lines indicate the basal side). **d**, Four phases during the invagination process. **e**, Multi-photon optical section near the hinge at phase 4. **f**, **g**, Effects of Y-27632 treatment (**f**, starting from phase 2 (ph2); **g**, from late phase 3 (ph3⁺)) on invagination. **h**, **i**, Treatment of the phase 3⁺ neural retina (blue) with (**i**) or without (**h**) aphidicolin. Red, 15 h later. Scale bars: 100 μ m (**c**, **h**), 20 μ m (**e**).

not shown; invagination 0%, $n = 30$, both phases). A similar effect was seen with the myosin inhibitor blebbistatin (data not shown). In contrast, when Y-27632 was added after the hinge appeared, the neural retina reproducibly continued to invaginate at phase 4 ($40 \pm 3.9\%$, $n = 30$, three experiments; Fig. 2g), indicating a phase-specific dependence (phases 1–3) of morphogenesis on ROCK activity.

At phase 4, the invaginating neural retina epithelium expanded substantially both in the tangential and vertical directions, whereas the RPE domain (particularly in its distal part) expanded exclusively tangentially (Supplementary Fig. 2e–g). Treatment with the mitotic inhibitor aphidicolin²⁹ from late phase 3 onwards efficiently blocked the invagination progress during phase 4 (Fig. 2h, i and Supplementary Movie 3) as well as tissue expansion and cell proliferation (Supplementary Fig. 2h–j), indicating that tissue expansion has an essential role in the phase-4 morphogenesis.

Stepwise acquisition of domain-specific properties

Detailed 3D analysis showed that, whereas the entire phase-1 vesicle consisted of simple columnar epithelium, the epithelial cells of the RPE, neural retina and hinge domains at phase 4 exhibited distinct morphologies, reminiscent of their shapes *in vivo*²⁶: columnar epithelial cells, pseudo-stratified cells, and apically narrow wedge-shaped cells, respectively (Fig. 3a, Supplementary Fig. 2d, bottom, and Supplementary Movie 2, part d; no wedge-shaped cells appeared at the hinge in culture pre-treated with either Y-27632 or blebbistatin; data not shown).

Both *in vitro* and *in vivo*, neural retina invagination occurs via unique apically convex invagination, in contrast to the well studied apically concave invagination seen in vertebrate neural-tube formation

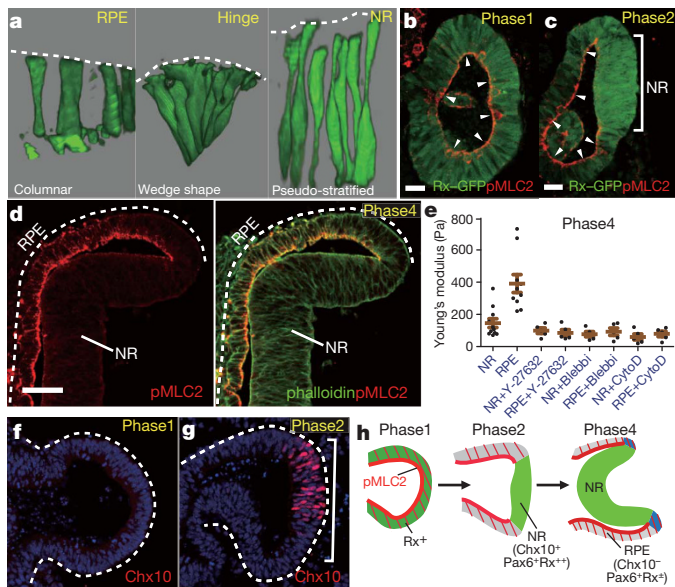


Figure 3 | Stepwise acquisition of domain-specific epithelial properties. **a**, Three-dimensional reconstruction views of the RPE, hinge and neural retina cells at phase 4. **b**, High pMLC2 levels (arrowheads) were seen in the entire Rx-GFP⁺ epithelium at phase 1. **c**, Reduced pMLC2 accumulation in the neural retina (bracket) at phase 2. **d**, Phase 4. Reduced and high pMLC levels in the neural retina and in the RPE and hinge regions, respectively. **e**, AFM-based measurement of relative tissue rigidity of phase 4 neural retina and RPE tissues with or without Y-27632, blebbistatin and cytochalasin D treatments. **f, g**, Expression of Chx10 at phase 1 (**f**) and phase 2 (**g**). **h**, Schematic summary. Scale bars: 50 μ m (**b–d**). Error bars in **e** represent s.e.m.

and during fly gastrulation, where ROCK-myosin activation is essential^{27,28,30–34}. With this in mind, we next examined levels of phosphorylated myosin light chain 2 (pMLC2; MLC2 is also called Myl2), which reflects local actomyosin activation³⁰, in different domains and at different phases. The proximal and distal epithelia of the phase-1 vesicle showed comparable levels of pMLC2 (Fig. 3b and Supplementary Fig. 3a). This pMLC2 accumulation seemed to depend on the Rho-ROCK system, as Y-27632 diminished pMLC2 levels (data not shown; consistent with this, the inner space of the phase-1 vesicle gradually decreased whereas Y-27632 reversed this reduction and expanded the whole vesicle evenly; Supplementary Fig. 3b). In contrast to the even accumulation at phase 1, the pMLC2 levels at phases 2–3 were reduced in the neural retina but remained high in the RPE and hinge epithelium (Fig. 3c and Supplementary Fig. 3c). The pMLC2 levels of the neural retina became particularly low at phase 4 (Fig. 3d and Supplementary Fig. 3d). The reduced pMLC2 level was also observed in the neural retina of the mouse embryo (Supplementary Fig. 3e, f). Consistent with the reduced level of pMLC, the mechanical rigidity of the RPE tissue in the phase-4 optic cup was substantially higher than that of the neural retina as measured by a force probe using an atomic force microscope (AFM) cantilever^{35,36}, whereas this difference disappeared after treatment with ROCK or actomyosin inhibitors (Fig. 3e and Supplementary Fig. 3g). In contrast, a substantial level of actomyosin-dependent tissue rigidity was seen in both the distal and proximal epithelia during phase 1 (Supplementary Fig. 3h and not shown).

We also observed differential domain- and phase-specific mechanical properties in response to cell ablation. When a few cells were ablated³⁷ by means of a 3D-targeted multi-photon laser beam (Supplementary Fig. 3i–l), the ablated space in the phase-4 RPE and neural retina, but not in the hinge, soon disappeared by compression from neighbouring cells within the epithelium (Supplementary Movie 4, parts a–c). Mitotic inhibitor treatment inhibited the gap-filling behaviour (Supplementary Movie 4, part d), indicating an essential role for

cell proliferation in generating the pushing force. In contrast, at phases 1–2, the ablation-induced gaps, whether in the distal or proximal epithelia, did not close after ablation but began to increase, presumably by local pulling tension (part e and not shown), as in the hinge during phase 4.

As for marker expression, Chx10 expression started by phase 2 in the distal retinal epithelium, whereas the proximal portion was Chx10[−] Pax6⁺ and showed decreased Rx expression (Fig. 3f–h; see *in vivo* expression in Supplementary Fig. 3m). Collectively, these findings demonstrate that the self-formed retinal epithelium sequentially acquires quantitatively distinct morphological, biochemical, mechanical and gene-expression properties in a domain-specific manner during early eye-cup morphogenesis.

Self-patterning into neural retina and RPE domains

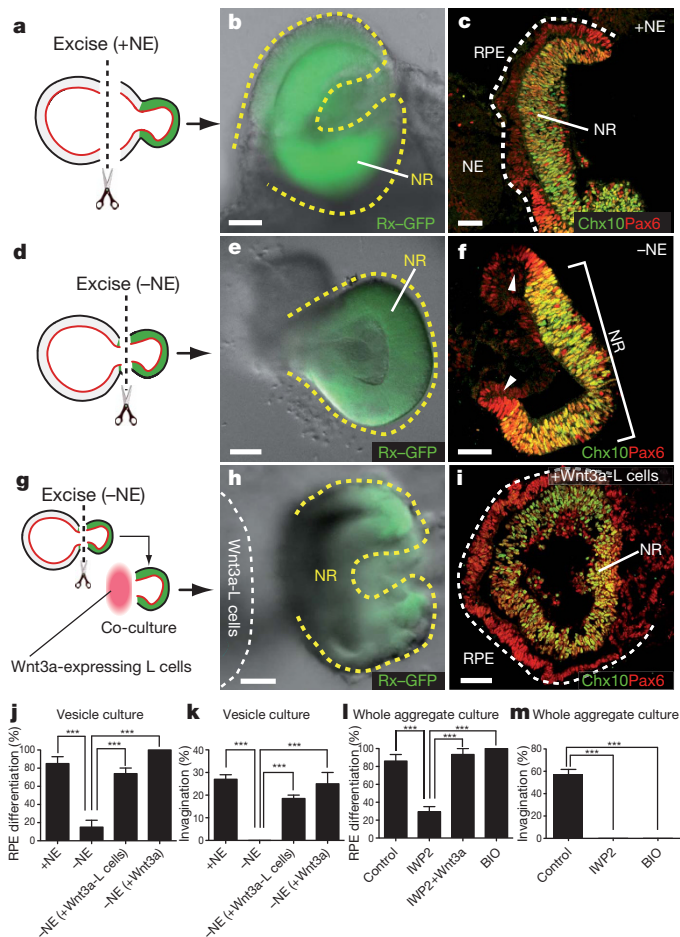
We next asked whether the neural retina and RPE fates were irreversibly determined when vesicle evagination was complete. To test this, we excised Rx⁺ phase-1 vesicles with or without neighbouring neuroectodermal epithelium tissue (Rx[−]) from the day-7 ES cell aggregates (Fig. 4a–k). A few days later, the Rx⁺ vesicles excised with a small amount of non-retinal neuroectodermal epithelium developed into spatially ordered RPE and neural retina epithelium, and even formed an optic-cup structure (Fig. 4a–c, j, k). In contrast, Rx⁺ epithelium isolated without non-retinal tissues failed to invaginate, and predominantly expressed Chx10 on day 10 (Fig. 4d–f; bracket), whereas only a small number of cells were Chx10[−] Pax6⁺ (arrowheads).

Canonical Wnt signalling has been implicated in the specification and maintenance of RPE fate^{38–41}. When phase-1 vesicles isolated without neuroectodermal epithelium tissue were co-cultured with an aggregate of Wnt3a-expressing L cells (Fig. 4g and Supplementary Fig. 4a, b), both the RPE differentiation and neural retina invagination were substantially rescued (Fig. 4h–k; Supplementary Movie 5, using a disc-confocal incubation microscope in Supplementary Fig. 2k), whereas such rescue was not observed with control L cells (not shown). A similar rescue was observed in the isolated vesicle culture to which Wnt3a protein was added (Fig. 4j, k and Supplementary Fig. 4c, d). Accordingly, treating the whole aggregates with the Wnt secretion inhibitor IWP2 interfered with invagination, reduced RPE tissues and increased the percentage of Chx10⁺ tissues (Fig. 4l, m and Supplementary Fig. 4e–h; the effects of IWP2 on differentiation were reversed by adding Wnt3a, whereas no substantial effects were seen on proliferation or apoptosis; Supplementary Fig. 4i, j). Conversely, the Wnt agonist BIO (6-bromoindirubin-3'-oxime; GSK3- β inhibitor) strongly promoted RPE differentiation and suppressed neural retina generation (Fig. 4l, m and Supplementary Fig. 4k, l).

Taken together, these findings indicate that RPE differentiation requires tissue interactions involving diffusible induction factors⁴² from neuroectodermal epithelium. In contrast, the neural retina fate is tissue-autonomously determined; this might, in part, result from Rx and Six3, which attenuate Wnt pathway activation⁴⁰, and also from Wnt antagonists such as Dkk1 (ref. 43), which is strongly expressed in the retinal tissues (Supplementary Fig. 4m–q). Interestingly, even at phase 1, *Dkk1* expression in the vesicle already showed a biased distally high expression (Supplementary Fig. 4r), which may partly explain why the addition of soluble Wnt3a protein in the explant culture could also spatially rescue the RPE–neural retina pattern (Supplementary Fig. 4d).

Self-directed stratification of neural retina tissue

In the growing phase-4 neural retina, the progenitors underwent frequent cell divisions coupled with interkinetic nuclear migration (Fig. 5a, Supplementary Fig. 5a and Supplementary Movie 6, parts a, b), as seen *in vivo*⁴⁴. This led us to examine the formation of tissue architecture in isolated neural retina epithelia in long-term culture. When phase-4 eye cups were manually excised en bloc at the RPE domain from the aggregates, the RPE portions flipped over like a leaf



spring at the hinge and everted in an inside-out fashion (Fig. 5b, c, Supplementary Fig. 5a and Supplementary Movie 7; such eversion was not seen in the samples shown in Fig. 4g presumably because their cut ends spontaneously became sealed soon after). In contrast to RPE, the shape of neural retina did not substantially change, demonstrating the tissue's plasticity; in fact, this mechanical property was useful for subjecting the isolated phase-4 neural retina epithelia to further suspension culture in retina maturation medium⁴⁵ (Supplementary Fig. 5b).

Notably, 10–14 days later, these isolated tissues spontaneously formed large, continuous epithelial structures with clear stratification (Fig. 5d; typically 800–2,000 μm in diameter), reminiscent of the early postnatal retina consisting of multiple layers of distinct cellular components^{5,13} (Supplementary Fig. 5c). The ES-cell-derived stratified retinal tissues contained all of the major neural retina components (Fig. 5e–k), that is, photoreceptors (rhodopsin⁺, recoverin⁺; Fig. 5e, h, i and Supplementary Fig. 5d), ganglion cells (Brn3⁺Pax6⁺calretinin⁺; Fig. 5g, h, k and Supplementary Fig. 5e), bipolar cells (Chx10⁺Pax6⁺,

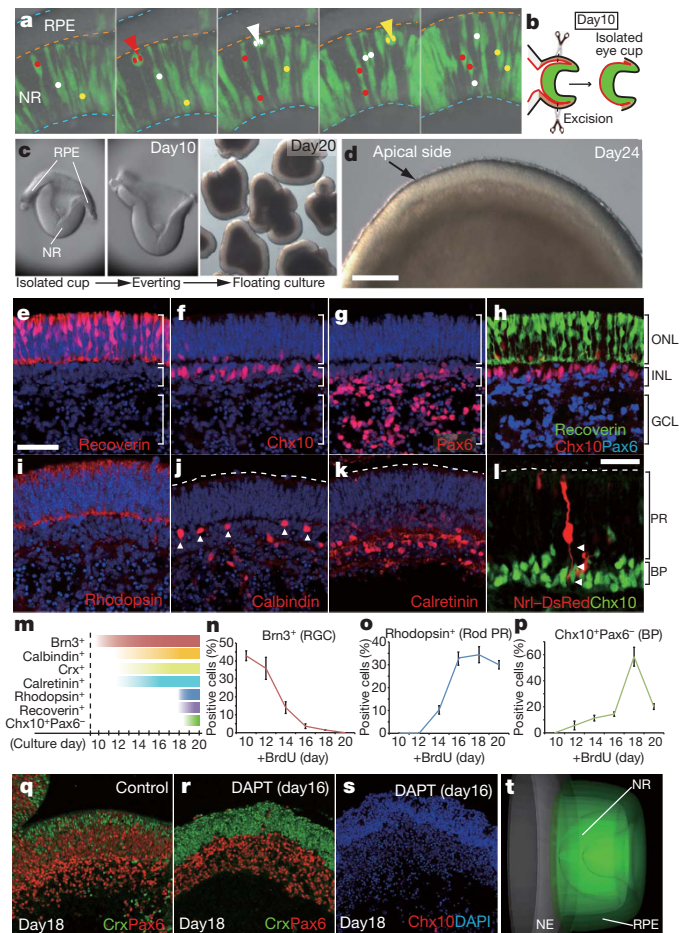


Fig. 5f–h; Chx10 expression is bipolar-cell-specific at this late stage), horizontal cells (calbindin⁺calretinin⁺; Fig. 5j), amacrine cells (Pax6⁺calretinin⁺; Fig. 5g, k) and Muller glia (CRALBP⁺; Supplementary Fig. 5f). These neural retina components are spatially arranged into the correct apical–basal order as seen in the early neonatal eye. Photoreceptors comprised the outermost layer, overlaying a layer of bipolar cells (Fig. 5h, l). In Fig. 5l, the shape of a photoreceptor cell is visualized with DsRed driven by the photoreceptor-specific *Nrl* promoter⁴⁶ (Supplementary Fig. 5g–i; *Nrl* expression started just before the terminal cell division of photoreceptors and became strong soon after; Supplementary Movie 6, part c). Horizontal cells were found between the photoreceptor and bipolar layers. The innermost zone contained ganglion cells and amacrine cells, and the formation of an innerplexiform-layer-like zone was seen there⁴⁷ (Supplementary Fig. 5j, arrow, and k).

As a minor difference, the ganglion cells in this culture tended to scatter rather than aligning as *in vivo*, probably because there was no

rigid border structure such as that provided by the inner limiting membrane *in vivo*. In addition, unlike rods, cone photoreceptors (expressing colour opsins) were present in a relatively low percentage (<0.1%; Supplementary Fig. 5l, m and data not shown).

The expression profile of stage-specific differentiation markers of neural retina components, summarized in Fig. 5m, indicated that the temporal order in ES-cell culture followed that *in vivo*. In generation-date analysis²⁰, the cell-cycle exit of ganglion cells (Brn3⁺), photoreceptors (rhodopsin⁺) and bipolar cells (Chx10⁺Pax6⁻) peaked around days 10, 16 and 18, respectively, which is in accordance with the *in vivo* generation order¹³ (Fig. 5n–p and Supplementary Fig. 5n, o). Consistent with these observations, treatment with the Notch inhibitor DAPT (on day 16), which promotes on-going differentiation and inhibits proliferation, increased the number of Crx⁺ photoreceptors on day 18 whereas it markedly reduced the number of Chx10⁺ cells (mitotic neural retina progenitors and postmitotic bipolar cells) and Ptf1a⁺ GABAergic precursors (Fig. 5q–s and Supplementary Fig. 5p–s, and data not shown).

Collectively, these findings demonstrate that the fully stratified neural retina tissue architecture in this ES-cell culture self-forms in a spatiotemporally regulated manner mimicking *in vivo* development.

Discussion

The self-directed organization of the complex optic-cup and neural retina morphology, shown here, was unanticipated, as the ES-cell culture started as patternless aggregates of homogeneous pluripotent cells, and continued under a uniform culture environment. Thus, this system offers a representation of the emergence of collective cell behaviour in a non-equilibrium biological process that evolves spatiotemporally. The invagination of the neural retina could occur even without forces from external structures at least in this *in vitro* context. Whereas the complete elucidation of the mechanics of this autonomous invagination requires further extensive investigation, this study indicates that the morphogenesis includes multiple steps controlled in a history-dependent fashion.

Following the formation of the hinge, phase-3 neural retina tends to have a slight apical bending. A working model for future investigation, consistent with our findings for tissue movement and mechanical properties during phase 4, is that, surrounded by the less-yielding RPE shell (pMLC2-high) and in a limited space, the flexible phase-4 neural retina is promoted to bend further (buckle apically) by its tangential expansion. Indeed, such a morphogenetic process could also be recapitulated by computer simulation based on the following three sequential local rules consistent with our observations: (1) relaxation of the presumptive neural retina beginning at phase 2; (2) apical constriction in the hinge epithelium at phase 3; and (3) rapid tangential growth of the neural retina and the distal RPE tissues at phase 4. In this model involving the local control of apical, basal and transmembrane springs, the neural retina spontaneously involutes inside the RPE shell, following flattening and hinge formation, conceptually supporting the feasibility of spontaneous cup morphogenesis independent of forces from external structures (Fig. 5t, Supplementary Fig. 6a, b and Supplementary Movie 8; see also Supplementary Fig. 6c–e for response to perturbations).

This study has revealed that the complex morphogenesis of the retinal anlage, at least in the *in vitro* context, possesses a 'latent intrinsic order' involving dynamic self-patterning and self-formation driven by a sequential combination of local rules and internal forces within the epithelium. The *in vivo* situation should be certainly more complex, and extrinsic signals and forces from external structures (for example, the surface ectoderm, lens and periocular mesenchyme)^{9,10,42,48,49} as well as space constraints presumably work together with this intrinsic order to reinforce robust retinal morphogenesis.

With regard to the application aspect, self-formation of fully stratified 3D neural retina tissues heralds the next-generation of generative medicine in retinal degeneration therapeutics, and opens up new

avenues for the transplantation of artificial retinal tissue sheets⁵⁰, rather than simple cell grafting.

METHODS SUMMARY

The SFEBq culture was performed as described previously²⁰, and matrigel (final 2%) was added to the 1.5% KSR-containing differentiation medium on day 1. Multi-photon imaging and ablation were performed using 900-nm femtosecond laser pulses (80 MHz) from a MaiTai eHP (Spectra-Physics) using a $\times 25$ water-immersion objective lens (NA 1.05; Olympus).

Full Methods and any associated references are available in the online version of the paper at www.nature.com/nature.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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METHODS

ES cell culture. Mouse ES cells (EB5, *Rx-Venus*, *Sox1-GFP*) were maintained as described²⁰. For the pigmentation assay (Fig. 1k), an ES cell line with B6/129 background was used. For SFEBq culture, ES cells were dissociated to single cells in 0.25% trypsin-EDTA and quickly re-aggregated in differentiation medium (3,000 cells per 100 μ l per well) in 96-well low-cell-adhesion plates (Lipidure Coat, NOF). Differentiation medium was G-MEM supplemented with 1.5% knockout serum replacement (KSR; Invitrogen) + 0.1 mM non-essential amino acids (Invitrogen) + 1 mM pyruvate + 1 mM 2-mercaptoethanol. As shown in Supplementary Fig. 1d, the addition of 1.5% KSR was much more efficient in Rx induction than that of 10% KSR, a condition that had been used previously²². Defining the day on which the SFEBq culture was started as day 0, matrigel (growth-factor-reduced; BD Biosciences) was added to culture to final 2% (v/v) on day 1. Alternatively, purified laminin and entactin proteins (200 μ g ml⁻¹; BD Biosciences) and nodal protein (1 μ g ml⁻¹) were added on day 1. The culture was transferred to bacterial-grade plastic dishes on day 7 and further cultured in suspension in DMEM/F12 medium supplemented with the N2 additive under 40%-O₂/5%-CO₂ conditions. The Rx induction occurred without substantial differentiation of mesodermal or endodermal tissues. For the sake of simplicity and cost, the matrigel treatment was used in the experiments unless otherwise mentioned.

Signalling molecules and antagonists. Growth factors were purchased from R&D Systems and used at the concentration indicated in the text and legends. The Notch inhibitor DAPT was applied to the isolated neural retina tissues from day 16. On day 18, the cell aggregates were fixed and analysed by immunohistochemistry. The integrin-blocking experiment (Supplementary Fig. 1) was performed by treating ES cells with the anti- β 1-integrin antibody or control IgM (10 μ g ml⁻¹; BD Pharmingen) from day 1, before the addition of matrigel to the culture.

IWP2 (4 μ M) and BIO (500 nM) also inhibited the neural retina invagination in the whole-head tissue culture of E8.75 mouse embryos. Consistent with the data shown in Fig. 4, they increased and decreased the amount of neural retina tissues in the eyes, respectively. In this case, however, these inhibitors also blocked the invagination and differentiation of the lens, demonstrating the limitation of the embryonic tissue explant assays and the advantage of the ES-cell culture system in terms of the simplicity in investigating the self-organizing nature of neural retina invagination.

Live imaging of optic-cup formation. The 3D live imaging was performed using specially assembled inverted microscopes (confocal or multi-photon) combined with a full-sized CO₂/O₂ incubator. The position of the ES cell aggregate or isolated vesicle was fixed in a drop of undiluted matrigel, which was then immersed in culture medium (described above) on a 3.5-cm glass-bottom dish. For confocal analysis, optical section images were obtained using a $\times$ 20 objective lens (Olympus), a spinning disk confocal system (CSU-X1, Yokogawa) and an EM-CCD camera (Andor, 512 $\times$ 512 pixels; the incubation system is based on LCV-110, Olympus, but the optical system including bilateral telecentric lenses was newly designed; for obtaining the whole aggregate view, a $\times$ 0.5 lens was interposed in the light path; Supplementary Fig. 2f). For high-resolution multi-photon live imaging, a stack of optical section images (512 $\times$ 512 pixels for the X–Y plane and 2 μ m for Z-axis step; typically 150 sections) was taken at each time point using a $\times$ 25 water-immersion lens (N.A. 1.05, Olympus) and multi-photon femtosecond laser (900 nm; Mai-Tai DeepSee eHP, Spectra-Physics) with group velocity dispersion auto-compensation (see Supplementary Fig. 2a, b). The 3D image analyses were done with Imaris 7.1 (Bitplane). The bright-field view in parallel to multi-photon imaging was obtained by scanning with a 900-nm femtosecond laser and detecting the transmission light with a separate photo-multiplier. Live imaging detected no substantial tangential migration or mixing of cells across different domains (non-retinal neuroectoderm, RPE and neural retina) of the ES-cell-derived epithelium, at least after day 3 (data not shown), indicating that distinct retinal domains are likely to be formed by region-specific patterning in the retinal epithelium rather than by the sorting of distinct cell types, which has been indicated as the mechanism in zebrafish retinogenesis⁷.

For membrane staining, cells were stained with the vital fluorescent dye Cell Mask. For high-resolution cell-shape images and cell-tracking images (Figs 3a and 5a), cells were partially labelled by co-aggregating *Rx-GFP* and non-labelled ES cells on day 0.

Culture of isolated optic vesicles and neural retina. For excision assays, *Rx-GFP*⁺ optic vesicles were mechanically excised (with or without non-retinal neuroectodermal epithelium tissues) from the main body of day 7 SFEB aggregates using no. 5 forceps, and were subjected to suspension culture under 40%-O₂/5%-CO₂ conditions in DMEM/F12 medium supplemented with N2. For the explant samples with neuroectodermal epithelium, the culture included approximately the same amount of neuroectodermal epithelium tissue with *Rx*⁺ vesicle. For co-culture assays, Wnt3a-expressing or control L cells (both lines were purchased from ATCC) were spun down in 1.5-ml tubes and the cell pellets were entrapped in a droplet of undiluted matrigel and subjected to co-culture with the isolated vesicles

without neuroectodermal epithelium tissues. In this co-culture, a small gap (100–150 μ m) was placed between an L cell aggregate and an isolated vesicle to minimize physical interactions. No substantial differences in mitosis (pH3⁺) or apoptosis (active caspase 3⁺) were seen between the proximal epithelia of the excised and non-excised vesicles. Unlike the cup in Fig. 5c, the optic cup forming from the isolated vesicles did not evert, presumably because the proximal end of the RPE spontaneously sealed and became immobilized itself.

For long-term neural retina culture, the neural retina portions of day 10 optic cups with some RPE were isolated mechanically using forceps and were subjected to suspension culture under 40%-O₂/5%-CO₂ conditions in DMEM/F12 medium supplemented with N2 + 10% FBS + 0.5 μ M all-*trans* retinoic acid (RA) + 1 mM L-taurine. The attaching RPE tissue everted or rolled up at one end of the growing neural retina vesicle during suspension culture. The culture grew and survived better with the 40% O₂ condition than with the 20% air. In particular, the survival of ganglion cells, which are first-born neurons, was much better supported by the high-O₂ condition. RA also improved the tissue growth. In the absence of RA, the photoreceptor layer tended to show some disorder in tissue architecture and contained some rosettes. In contrast, the absence of L-taurine had only a marginal effect in this culture. The neural retina culture could be continued until day 35, and, beyond this, the tissue gradually lost its integrity (in future investigation, it is necessary to establish improved conditions for longer culture to induce the further maturation of functional photoreceptors, for example, in the development of outer segments).

Electroporation of the *pNrl-DsRed2* plasmid (from Addgene) with the *pCAG-H2B-GFP* plasmid was performed on the apical side of day-16 neural retina tissues using the CUY21 pulse generator (BEX; 30 V, 50 ms $\times$ 5 pulses, 450-ms interval) and a pair of platinum electrodes (2.5-mm gap; BEX). On days 18–20, 20–30% of H2B-GFP⁺ cells co-expressed DsRed2.

Three-dimensional multi-photon laser ablation and mitosis inhibition.

Three-dimensional laser ablation was done using the same multi-photon optical system for 3D live imaging. 900-nm laser beam from Mai-Tai Deep-See eHP at full power was condensed at the target cells via a $\times$ 25 water-immersion lens (N.A. 1.05) using a spot illumination or line scan until cell rupture (typically 0.4–2 s duration). The gap created by cell rupture was quantified as a maximal gapped area in the optical section at each time point. When the gap was closed, the content of ruptured cells (debris) was usually pushed out from the epithelium via the apical side, probably because the other side was blocked by the basement membrane. Although the created gaps remained open in the tissues with pulling tension (for example, the epithelia at the early phases) for a substantial period after ablation, they eventually started closing, for example, 1 h after, probably reflecting a secondary wound healing.

The mitotic inhibitor aphidicolin was used to attenuate the proliferation during phase 4 by adding it at a final concentration of 5 μ M to culture at the end of phase 3 and incubating the culture for 5 h before analysis (its effects on cell proliferation and tissue expansion are shown in Supplementary Fig. 2h–j).

Immunohistochemistry, qPCR and FACS. Immunohistochemistry was performed as described previously²⁰. Antibodies against the following proteins were used at the indicated dilutions: Nanog (rabbit/1:500, Reprocell), Oct3 (mouse/1:500, Transduction Laboratories), N-cadherin (mouse/1:1,000, BD Transduction), Pax6 (mouse/1:1,000, R&D), Chx10 (sheep/1:1,000, Exalpha), Rx (rabbit/1:1,000¹⁵), laminin (rat/1:500, Chemicon, rabbit/1:500, Abcam), PKC ζ (rabbit/1:200, Santa Cruz), CD133 (rat/1:500, Chemicon), Mitf (mouse/1:1,000, Exalpha), Coup-TF2 (mouse/1:1,000, Perseus Proteomics), connexin 43 (rabbit/1:1,000, Sigma), recoverin (rabbit/1:1,000, Chemicon), Otx2 (goat/1:100, Santa Cruz), Brn3 (goat/1:50, Santa Cruz, C-13), rhodopsin (mouse/1:1,000, Sigma), calbindin (mouse/1:1,000, Swant), calretinin (rabbit/1:1,000, Swant), CRALBP (also called Rlbp1; mouse/1:500, Abcam), VGLUT1 (rabbit/1:1,000, Wako), bassoon (mouse/1:200, Assay Designs), pMLC2 (rabbit/1:500, Cell Signaling), pH3 (mouse/1:500, Cell Signaling). DAPI was used for counterstaining the nuclei (Molecular Probes). Stained sections were analysed with an LSM710 confocal microscope (Zeiss).

Quantitative PCR was performed using the 7500 Fast Real Time PCR System (Applied Biosystems) and data were normalized to GAPDH expression. Primers used were as follows: *Dkk1*, forward 5'-CCGGGAAGTACTGCAAAAT-3', reverse 5'-CCAAGGTTTCAATGATGCTT-3'; *Dkk3*, forward 5'-TCGTGAC CAGATCCAGCTT-3', reverse 5'-AGCGCTGCATGTTTGT-3'; *Rx*, forward 5'-CGACGTTCCACCTTACCA-3', reverse 5'-TCGGTCTGGAACCATA CCT-3'; *Six3*, forward 5'-CCGGAAGAGTTGTCCATGTTC-3', reverse 5'-CG ACTCGTGTGTTGTGATGGC-3'; *Ncad*, forward 5'-CAGGTTGGACGTCAT TGATG-3', reverse 5'-AGGTCCTCCACCATGATTC-3'. The values shown on graphs represent the mean $\pm$ s.e.m. For FACS analysis, cells were counted with FACSaria (Becton Dickinson), and the data were analysed with the FACSDiva software (Becton Dickinson).

Birth-dating analysis. For the *in vitro* birth-dating analysis, aggregates were treated with BrdU (5 μ g ml⁻¹, Invitrogen) on days 10, 12, 14, 16, 18 or 20, and

rinsed with medium 12 h after, to remove the BrdU. On day 24, the cell aggregates were fixed and cryosectioned. Sections were immunostained and the percentages of BrdU⁺ cells that were positive for each lineage marker were quantified. For quantification, 16–24 aggregates were examined for each experiment, which was repeated at least three times.

Tissue elasticity measurement. The tissue elasticity/rigidity of ES-cell-derived retinal epithelia was examined by indentation assay using an AFM cantilever as described previously^{35,36}. Force probe was prepared by attaching a glass bead (100 µm diameter) to a tipless silicon cantilever (450 µm long, 50 µm wide, 2 µm thick; nominal spring constant $\sim 0.02 \text{ N m}^{-1}$; TL-CONT, Nanosensors) using two-component Araldite epoxy glue (the spring constant of the cantilever with bead was typically $\sim 0.1 \text{ N m}^{-1}$). The measurement using a contact mode was carried out with the Nano-Wizard system with the Cell-Hesion module (JPK), which is optimized for biological samples. Before the measurement, the spring constant of the force probe was calibrated according to the manufacturer's instruction by the thermal fluctuation method. The retinal tissues were pressed with a force of 20 nN (the approach and retraction speeds were set to $10\text{--}20 \text{ µm s}^{-1}$) by the force probe and the data analysis was done with the JPK image processing software, in which the tissue rigidity is calculated as a value of pushing stress (Pa) necessary for unitary strain of the tissue (Young's modulus determined by a force–distance curve). The measurement was done on the apical side of ES-cell-derived retinal tissues in the culture medium (DMEM/F12/N2) on top of a poly-D-lysine-coated plastic culture dish (several points in each RPE or neural retina epithelium). The inhibitor treatments were given for 60 min at 37°C before the measurements. All measurements were performed at room temperature.

Computer-simulated animations for the working model. A two-dimensional continuum model of epithelium consisting of multiple cells along the tangential direction was discretized into quadrilateral elements expressed by their vertex coordinates $\mathbf{r}_i$ as shown in Supplementary Fig. 6b. The dynamic formation process of the optic cup structure was numerically demonstrated under the axisymmetric condition to reduce the geometric complexity.

The vertices were connected by linear springs representing elastic behaviours of the element sides, and their motions were determined to decrease the total potential energy of the system U :

$$U = U_e + U_v + U_s \quad (1)$$

by solving the overdamped equation of motion^{51–53}:

$$\eta \frac{d\mathbf{r}_i}{dt} = - \frac{\partial U}{\partial \mathbf{r}_i} \quad (2)$$

where η represents the viscosity of the system. The energy U_e is elastic strain energy of the springs expressed as

$$U_e = \sum \frac{1}{2} k_e^\alpha (l^\alpha - l_{eq}^\alpha)^2 \quad (3)$$

where l^α is the length of element side, l_{eq}^α is its natural length at the stress-free state, and k_e^α is the elastic constant for $\alpha = a$ (apical), b (basal), t (transmural). The energy U_v for volumetric change of the elements

$$U_v = \sum \frac{1}{2} k_v (V - V_{eq})^2 \quad (4)$$

was introduced to take into account the local volumetric constraint of the elements, where V is the element volume, V_{eq} is the equilibrium element volume,

and k_v is the volumetric elastic constant. The energy U_s was introduced to express the resistance to the distortion of the elements (the 'self-righting' effect), as

$$U_s = \sum \frac{1}{2} k_s \{1 - \cos(\theta_1^\alpha - \theta_2^\alpha)\} \quad (5)$$

where θ_1^α and θ_2^α are the angles of element corners for $\alpha = a$ (apical), b (basal), and t is the elastic constant.

A hemispherical model of the epithelium vesicle with a constant thickness in the transmural direction was initially created in the stress-free state as a reference configuration for the elastic energies, and the epithelium, from the neuroectodermal epithelium part to the centre of the neural retina, was equally discretized into 100 elements. Deformation of the epithelium in the r – z plane generated by the hoop stress (circumferential stress), which comes from the three-dimensionality (curvature in the circumferential direction) of the vesicle under the axisymmetric condition, was considered by applying apparent force to each vertex in the radial direction that was assumed to be simply proportional to the magnitude of radial displacement of the vertex referring to its stress-free configuration. Then, a half domain of the hemispherical epithelium vesicle was analysed by assuming the symmetry with respect to the z axis as shown in Supplementary Fig. 6b.

The computer-simulated animations were produced on the basis of the following assumption deduced from the experimental results of the present study: (1) relaxation of the presumptive neural retina beginning at phase 2; (2) strong apical constriction of the hinge domain at phase 3; and (3) rapid tangential growth of the distal RPE and (tangential/vertical) growth of the neural retina at phase 4.

Phase 1: to create the initial condition, apical constriction in the retinal epithelium was simulated by reducing the natural length of the elements l_{eq}^a , which results in the formation of the evaginating vesicular shape.

Phase 2: elastic strain energy U_e stored in the tangential elements on the apical and basal surfaces in the presumptive neural retina was gradually released by changing their natural length, l_{eq}^a and l_{eq}^b , to the relaxed configurations with decreased elastic constants k_v and k_s . Simultaneously, equilibrium element volume V_{eq} in the neural retina region was relaxed to the corresponding stress-free state.

Phase 3: strong apical constriction in the hinge epithelium was simulated by reducing the natural length l_{eq}^a with an increase in the elastic constant k_e^a at the corresponding region. In the neural retina, the apical and basal elastic constants (k_e^a and k_e^b) were gradually decreased until the end of phase 3 (by 75% and also kept at this level during phase 4).

Phase 4: tangential expansion of the distal RPE and the neural retina at constant growth rates was simulated by increasing the size (natural length and equilibrium volume) of elements in the corresponding directions. To mimic the *in vitro* morphology, vertical expansion was also introduced in the central portion of the neural retina during phases 3–4 (although this did not show a strong impact on the neural retina invagination *in silico*).

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